PROJECT ABSTRACT

AI-Powered Financial Risk Prediction System

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Project Title	AI-Powered Financial Risk Prediction System
Academic Year	Third Year Final Project
Project Scope	Live / Secondary Data-Driven Predictive Machine Learning System

1. ABOUT THE PROJECT

In modern volatile financial markets, corporate stability relies heavily on real-time risk assessment and proactive distress forecasting. Traditional risk evaluation methods suffer from manual audit delays, outdated quarterly metrics, and rigid rule-based scoring models. The **AI-Powered Financial Risk Prediction System** addresses these limitations by providing an automated, machine-learning-driven analytics framework designed to analyze multi-dimensional transactional and market data, enabling real-time default prediction, fraud detection, and explainable credit risk scoring.

2. KEY OBJECTIVES

- **Automate Credit Default Forecasting:** Train supervised learning models (XGBoost, Random Forest) on financial ratios to predict probability of default and insolvency.
- **Detect Fraud & Transaction Anomalies:** Deploy unsupervised models (Isolation Forests) to identify irregular financial behavior and operational risk patterns.
- **Deliver Explainable AI (XAI):** Integrate SHAP frameworks to provide transparent, interpretable risk metrics for financial analysts.
- **Interactive Risk Dashboard:** Build a real-time web portal to visualize risk scores, trends, and alert indicators for decision-makers.

3. MAIN TECHNOLOGIES USED

- **Machine Learning & AI:** Python, Scikit-Learn, XGBoost, SHAP (Explainable AI), Pandas, NumPy.
- **Backend & API Layer:** FastAPI / Flask (Python), REST APIs, SQLAlchemy.
- **Frontend & Dashboard:** React.js / Next.js, Tailwind CSS, Recharts / Chart.js for data visualization.
- **Database & Storage:** PostgreSQL (Structured Logs & Telemetry), Vector/File Storage.

4. EXPECTED BENEFITS

- **Early Distress Warning:** Identifies credit degradation and liquidity risks months before conventional audit cycles.
- **Enhanced Prediction Accuracy:** Minimizes false positives and credit default misclassifications using ensemble algorithms.
- **Transparent Decision-Making:** XAI features give clear reasoning behind risk scores, improving regulatory and audit compliance.
- **Operational Efficiency:** Reduces manual risk review processing time from days to automated real-time assessments.

Keywords: *Financial Risk Prediction, Machine Learning, Credit Default, Anomaly Detection, Explainable AI (SHAP), FastAPI, React.js.*